

2 Numeric Variables Guided Lab 2025

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9/2/25

Importing & Cleaning Data

```
library(readr)
Quarterbacks <- readr::read_csv("C:/Users/RobTop/Documents/STA227 - Data Visualization/Quarterbacks.csv")

## Rows: 251 Columns: 20
## -- Column specification -----
## Delimiter: ","
## chr (2): Player, HOF
## dbl (18): HOFScore, ChampionshipWins, AllProSelections, ProBowlSelections, Y...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
Quarterbacks <- na.omit(Quarterbacks) #Remove 2 Rows With Errors or Missing value
head(Quarterbacks)

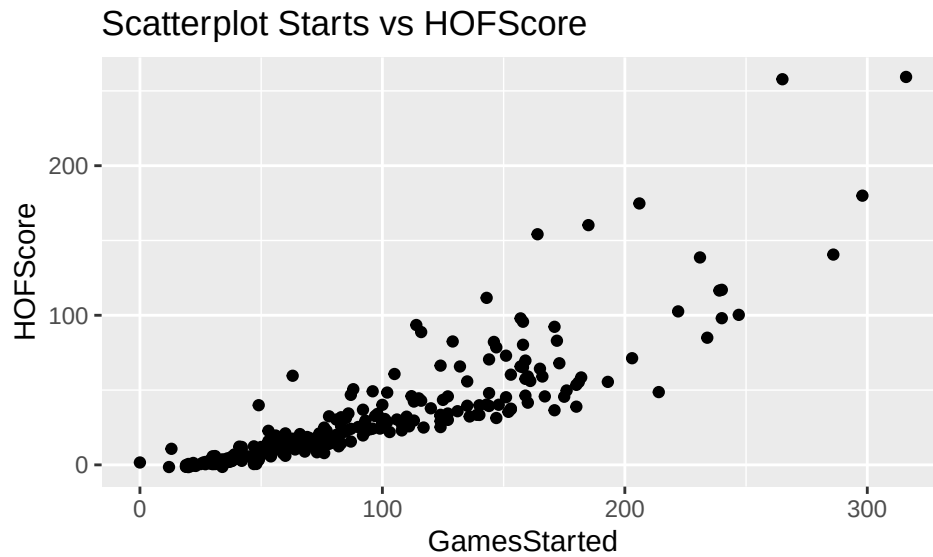
## # A tibble: 6 x 20
##   Player      HOFScore ChampionshipWins AllProSelections ProBowlSelections
##   <chr>      <dbl>          <dbl>          <dbl>          <dbl>
## 1 Tom Brady    259.              7              3              15
## 2 Peyton Manning 258.              2              7              14
## 3 Brett Favre   180.              1              3              11
## 4 Aaron Rodgers 175.              1              4              10
## 5 Johnny Unitas 160.              3              5              10
## 6 Joe Montana  154.              4              3              8
## # i 15 more variables: YearsStarter <dbl>, FirstYearStarter <dbl>,
## #   LastYearStarter <dbl>, WeightedCareerAve <dbl>, GamesPlayed <dbl>,
## #   GamesStarted <dbl>, PassesCompleted <dbl>, PassesAttempted <dbl>,
## #   YardsGained <dbl>, PassingTouchdowns <dbl>, LongestPass <dbl>,
## #   Interceptions <dbl>, Sacked <dbl>, YardsLostSacks <dbl>, HOF <chr>
```

Exploring Data

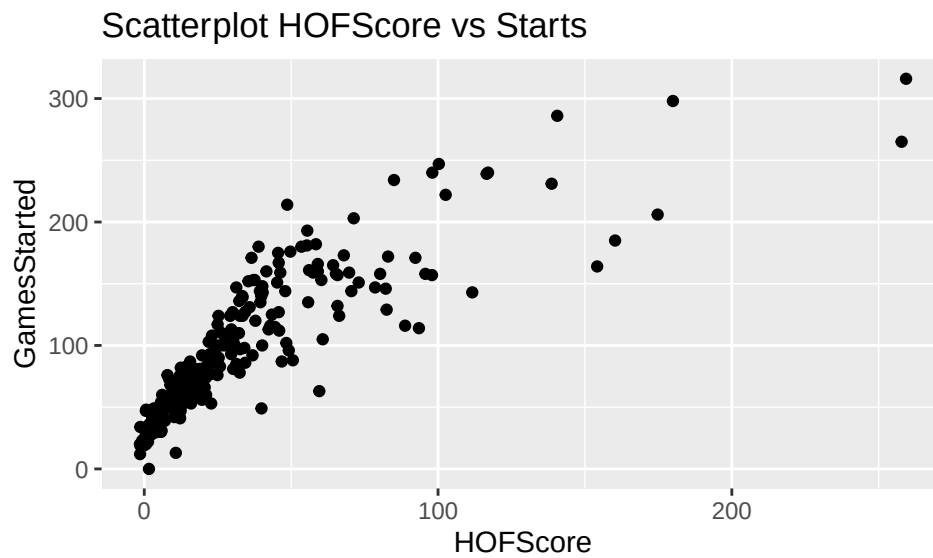
I would say theres probably a decent argument for those who play more have a higher score however players like Barry Sanders would likely have a similar score to some 'one hit wonder' players due to lack of games in comparison to a few big season players. Curious to see what happens.

Application

```
gf_point( HOFScore ~ GamesStarted , data = Quarterbacks, title="Scatterplot Starts vs HOFScore")
```

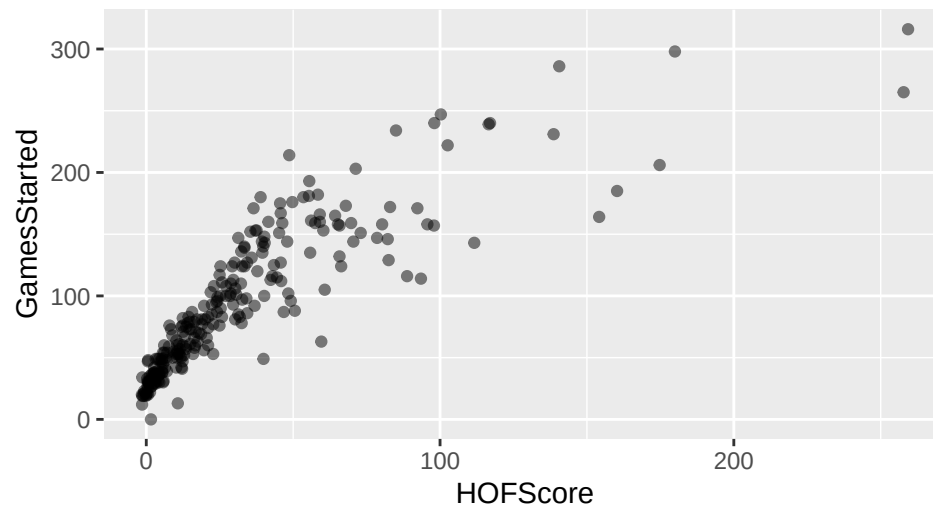


```
gf_point( GamesStarted ~ HOFScore , data = Quarterbacks, title="Scatterplot HOFScore vs Starts")
```



```
gf_point( GamesStarted ~ HOFScore , data = Quarterbacks , alpha = 0.5 , title="Scatterplot HOFScore vs Starts")
```

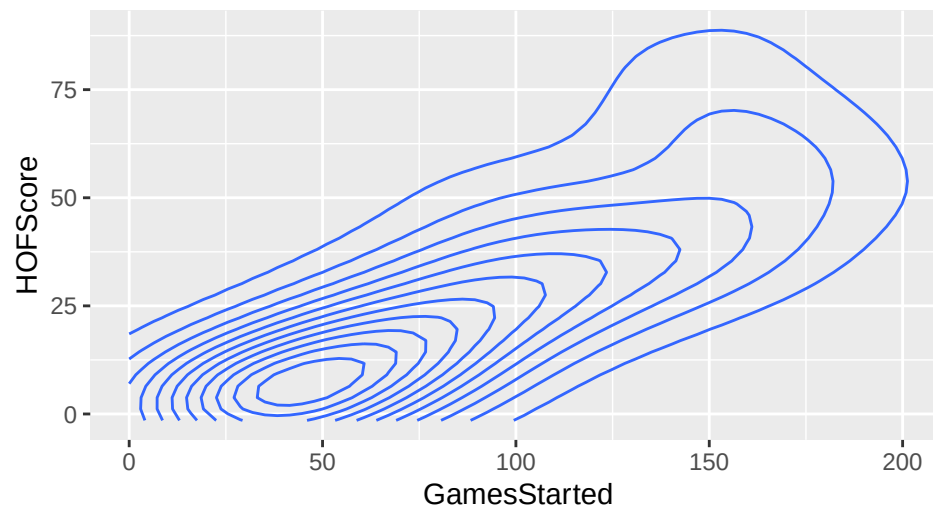
Scatterplot HOFScore vs Starts , alpha = 0.5



Contour Lines & Layers

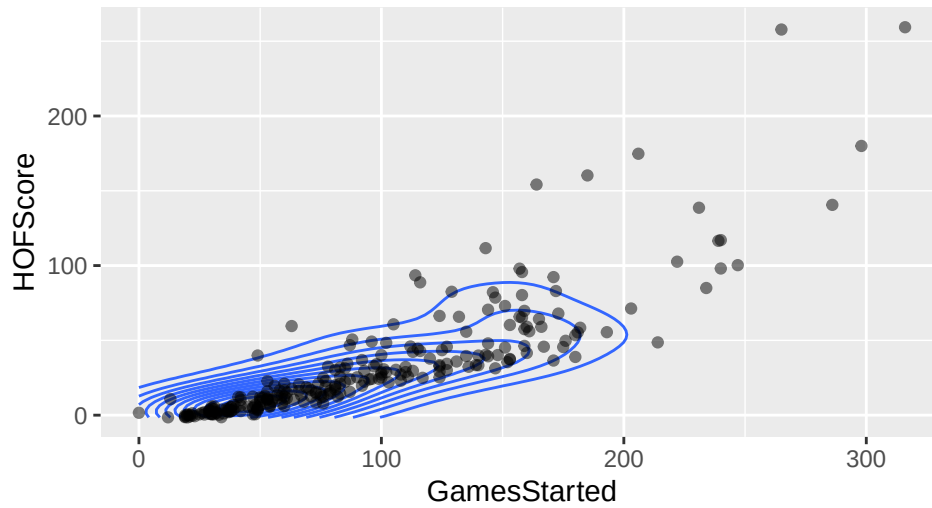
```
gf_density_2d( HOFScore ~ GamesStarted , data = Quarterbacks, title="gf_density_2d Starts vs HOFScore")
```

gf_density_2d Starts vs HOFScore



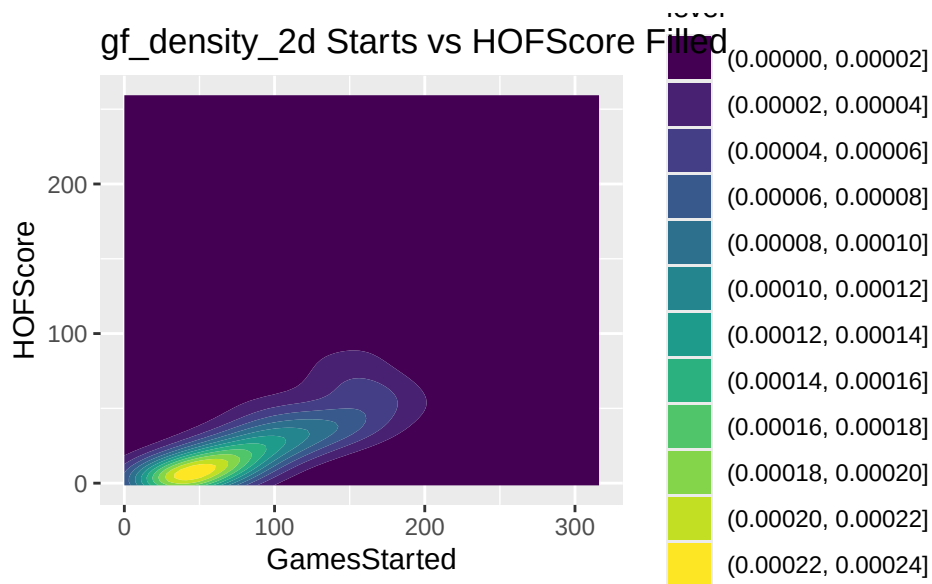
```
gf_density_2d( HOFScore ~ GamesStarted , data = Quarterbacks, title="contour lines and each individual")
```

contour lines and each individual point

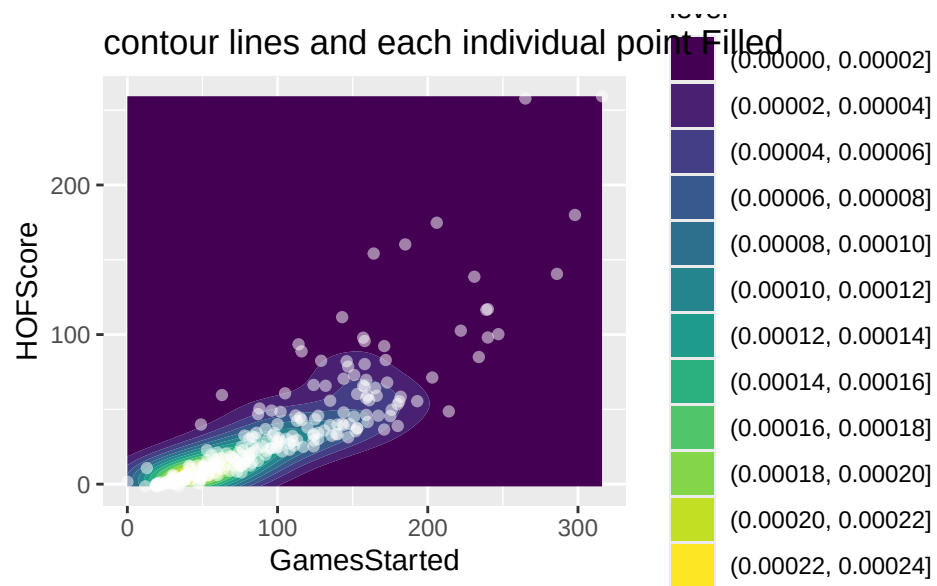


Density Filled (With Layers)

```
gf_density_2d_filled( HOF Score ~ GamesStarted , data = Quarterbacks, title="gf_density_2d Starts vs HOF Score Filled")
```



```
gf_density_2d_filled( HOF Score ~ GamesStarted , data = Quarterbacks, title="contour lines and each indi")
```



Hex Plot

```
gf_hex( HOFScore ~ GamesStarted , data = Quarterbacks, title="Hexplot Starts vs HOFScore")
```

